

Class Notes: Phylum Porifera

Animal Kingdom • PU-I Biology • Class Notes

Organisation and Water Canal System

Commonly called sponges; generally marine, primitive, multicellular, asymmetrical, with cellular level of organisation. Have a water transport / canal system: water enters through ostia into the spongocoel and leaves through the osculum, helping in food gathering, gas exchange and excretion.

Skeleton and Reproduction

Spongocoel and canals are lined by collar cells (choanocytes); digestion is intracellular. Skeleton consists of spicules or spongin fibres.

Examples

Reproduction: asexual (fragmentation, budding, gemmules) or sexual (gametes); sponges are hermaphrodite/monoecious with internal fertilisation and indirect development (amphiblastula larva in Sycon, parenchymula larva in Leucosolenia). Examples: Sycon, Spongilla (fresh water sponge), Euspongia (Bath sponge).